

# Speech Signal Analysis Using Praat

The word 'praat' means 'talk' or 'speak' in Dutch. This article introduces the reader to Praat, a rich, feature-laden piece of software that helps to analyse speech signals. It covers phonetics and various types of speech analyses.

Speech plays the most important role during verbal communication between humans. And phonetics is the study and classification of speech sounds. It is basically categorised into the following three groups:

1. Articulatory phonetics deals with speech production and how sound is produced.
2. Auditory phonetics deals with speech perception, and how sound is perceived.
3. Acoustic phonetics deals with the study of properties of the speech wave.

This process is termed the speech chain or the speech communication pathway.

## An introduction to speech processing

Analysis of speech signals is an important task. It deals with the manipulation of sound signals to extract meaningful information from them. There are many techniques for speech processing and, in this article, we will discuss some basic methods. Interested readers can go through an excellent tutorial on Praat at <http://www.fon.hum.uva.nl/david/sspbook/sspbook.pdf> (Reference 1).

Speech, being a non-stationary signal, continuously keeps on changing; hence, in order to model the speech signal, we follow the strategy of segmentation, which is the process of assuming the speech wave to be a static signal for a short period of time in which it remains almost constant. The typical length of such intervals is 20ms to 30ms. Hence, a speech signal is segmented into frames of 20ms-30ms, and each frame is analysed separately.

## Spectrum analysis

Normally, the signal is represented in the time domain, where the horizontal axis represents time and the vertical axis represents the amplitude (loudness) of the speech signal (i.e., amplitude  $\rightarrow$  time). You can look at the upper half of Figure 3, which is a time domain signal. The difficulty with the time domain is that it can provide only a limited amount of information about the signal. Fourier analysis is an important process, which transforms the signal from the time domain to the frequency domain, also referred to as the Fourier domain. The frequency domain represents the signal in amplitude (decibel)  $\rightarrow$  frequency (Hertz) dimensions. The signal representation in the frequency domain is also known as the frequency spectrum, or just the spectrum of the signal.

|        | 1209 Hz | 1336 Hz | 1477 Hz | 1633 Hz |
|--------|---------|---------|---------|---------|
| 697 Hz | 1       | 2       | 3       | A       |
| 770 Hz | 4       | 5       | 6       | B       |
| 852 Hz | 7       | 8       | 9       | C       |
| 941 Hz | *       | 0       | #       | D       |

Table 1: DTMF standards

A simple time domain signal (with a single frequency  $f$ ) shows a single spike at  $f$  Hz in the frequency domain. A composite signal (combination of multiple frequencies) presents quite a complex spectrum. Let us look at the following simple example—Dual Tone Multi Frequency